

VALUE ANALYSIS

Value Analysis is a systematic and organized approach used to improve the value of a product or service by analyzing its functions. It focuses on achieving the required function at the lowest possible cost without affecting quality, performance, or reliability.

It is mainly applied to existing products to reduce cost and improve efficiency.

A systematic method to improve the value of a product by analyzing its functions and eliminating unnecessary costs without reducing quality.

Value is expressed as:

Value=Function/cost

Objectives of Value Analysis:

1. Reduce cost without affecting quality
2. Improve product design and performance
3. Eliminate unnecessary features
4. Enhance customer satisfaction
5. Increase profitability
6. Promote efficient use of resources

Types of Functions

1. Basic Function

- Primary purpose of the product
- Example: A fan → to provide air

2. Secondary Function

- Additional or supporting features
- Example: Decorative design of fan

Steps in Value Analysis

1. Information Phase

- Collect all data about the product
- Understand design, cost, materials, and functions

2. Function Analysis Phase

- Identify and define functions
- Classify into basic and secondary functions

3. Creative Phase

- Generate alternative ideas
- Brainstorm ways to reduce cost or improve function

4. Evaluation Phase

- Analyze feasibility of ideas
- Select best alternatives

5. Development Phase

- Develop selected ideas into workable solutions

6. Implementation Phase

- Apply improvements in actual production

7. Follow-up Phase

- Monitor results and ensure effectiveness

Advantages of Value Analysis

- Reduces production cost
- Improves product quality
- Eliminates unnecessary features
- Encourages innovation
- Enhances customer satisfaction

Limitations

- Time-consuming process
- Requires skilled team
- Resistance to change
- Not suitable for very small projects

Example of Value Analysis

Example: Ball Pen

- Basic function → Writing
- Secondary function → Attractive design

Improvement:

- Replace expensive material with cheaper plastic
- Simplify design

Result:

- Reduced cost
- Same function maintained

VALUE ENGINEERING (VE)

Value Engineering (VE) is a systematic and organized approach used to improve the value of a product at the design stage. It focuses on achieving the required functions at the lowest possible cost without compromising quality, reliability, or performance.

Unlike Value Analysis (which is applied to existing products), Value Engineering is applied before production, during the design phase.

Value Engineering can be defined as:

A systematic application of techniques to identify functions of a product and provide those functions at the lowest cost consistent with required performance, reliability, and quality.

Objectives of Value Engineering

1. Reduce product cost at design stage
2. Improve product quality and performance
3. Eliminate unnecessary features
4. Optimize material and process selection
5. Increase customer satisfaction
6. Enhance profitability

Key Principles of Value Engineering

1. Focus on functions rather than components
2. Eliminate unnecessary costs
3. Encourage creativity and innovation
4. Use a multidisciplinary team approach
5. Maintain required quality and reliability

Phases / Steps in Value Engineering

1. Information Phase

- Collect all data about product design
- Understand specifications, materials, and costs

2. Function Analysis Phase

- Identify basic and secondary functions
- Define functions in simple terms

3. Creative Phase

- Generate alternative ideas using brainstorming
- Look for innovative solutions

4. Evaluation Phase

- Analyze alternatives
- Select best feasible option

5. Development Phase

- Develop selected idea into detailed design

6. Presentation Phase

- Present proposal to management
- Get approval for implementation

7. Implementation Phase

- Apply improvements in production

8. Follow-up Phase

- Monitor performance and results

Advantages of Value Engineering

- Reduces cost at early stage
- Improves product design
- Enhances quality and reliability
- Encourages innovation
- Saves materials and resources

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Example of Value Engineering

Example: Electric Fan Design

- Basic function → Provide air
- Secondary function → Attractive appearance

Improvement:

- Use lightweight and low-cost materials
- Simplify blade design
- Optimize motor efficiency

Result:

- Reduced manufacturing cost

- Improved performance

Basis **Value Analysis** **Value Engineering**

Stage **Existing product** **Design stage**

Purpose **Cost reduction** **Cost prevention**

Focus **Improve value** **Design optimization**

Implementation Procedure of Value Engineering (VE)

The implementation of Value Engineering involves a systematic and step-by-step procedure to improve product value by analyzing functions and reducing unnecessary costs during the design stage. It ensures that the required performance is achieved at minimum cost.

1. Information Phase

Activities:

- Collect all relevant information about the product
- Study design, specifications, materials, and cost
- Identify high-cost areas

Purpose: To clearly understand the product and its components.

2. Function Analysis Phase

Activities:

- Identify functions of each component
- Classify into:
 - Basic functions
 - Secondary functions
- Use tools like FAST diagram

Purpose: Focus on what the product must do, not how it does it.

3. Creative Phase

Activities:

- Generate alternative ideas
- Use brainstorming techniques
- Encourage innovative and cost-saving ideas

Purpose: Develop multiple solutions without criticism.

4. Evaluation Phase

Activities:

- Evaluate all alternatives
- Consider feasibility, cost, and performance
- Select the best option

Purpose: Choose the most effective and economical solution.

5. Development Phase

- Develop selected ideas into detailed design
- Prepare drawings, specifications, and cost estimates

Purpose: Convert ideas into practical solutions.

6. Presentation Phase**Activities:**

- Present proposals to management
- Explain benefits and cost savings
- Obtain approval

Purpose: Get management acceptance.

7. Implementation Phase**Activities:**

- Apply approved changes in production
- Modify processes and materials

Purpose: Execute the improved design.

8. Follow-up Phase**Activities:**

- Monitor performance
- Compare expected and actual results
- Ensure sustained improvement

Purpose: Maintain effectiveness of improvements

Example: Value Engineering of an Electric Fan**Problem Statement**

A company manufactures electric fans, but:

- Production cost is high

- Market competition is increasing
- Need to reduce cost without affecting performance

Step 1: Information Phase

- Product: Electric fan
- Cost breakdown:
 - Motor → High cost
 - Blades → Medium cost
 - Body → Medium cost

Identify **motor and material** as major cost contributors

Step 2: Function Analysis

Basic Function:

- Provide air (cooling effect)

Secondary Functions:

- Attractive design
- Noise reduction
- Safety

Focus on maintaining basic function at lower cost

Step 3: Creative Phase

Generate ideas:

- Use lightweight plastic blades instead of metal
- Improve motor efficiency
- Reduce unnecessary decorative parts
- Use standard components

Step 4: Evaluation Phase

Idea	Cost	Performance	Decision
Plastic blades	Low	Good	Selected
Remove design parts	Low	No effect	Selected
Cheaper motor	Low	Poor	Rejected

Select best cost-effective ideas

Step 5: Development Phase

- Design new fan with:

- Plastic blades
- Optimized motor
- Simplified structure

Step 6: Presentation Phase

- Present proposal to management
- Show:
 - Cost reduction
 - Same performance

Get approval

Step 7: Implementation Phase

- Start manufacturing new design
- Replace old materials

Step 8: Follow-up Phase

- Monitor performance
- Check customer satisfaction

Result:

- Cost reduced by 15–20%
- Performance maintained

Final Result

- Same function (air delivery)
- Reduced cost
- Increased profit
- Better competitiveness

ENTERPRISE RESOURCE PLANNING (ERP)

Enterprise Resource Planning (ERP) is an integrated management system that helps organizations plan, manage, and control all resources and business processes using a single unified software system.

ERP integrates different departments such as:

- Finance
- Human Resources
- Production
- Inventory
- Sales

This ensures smooth information flow and better decision-making.

ERP can be defined as:

A computer-based integrated system that manages and coordinates all organizational resources, information, and functions in real time.

Objectives of ERP

1. Integrate all business functions
2. Improve efficiency and productivity
3. Provide real-time information
4. Reduce operational costs
5. Enhance decision-making
6. Improve customer service

Components / Modules of ERP

Major Modules:

- 1. Finance Module**
 - Accounting, budgeting, cost control
- 2. Human Resource Module**
 - Recruitment, payroll, training
- 3. Production Module**
 - Planning, scheduling, quality control
- 4. Inventory Module**
 - Stock management, warehousing
- 5. Sales and Marketing Module**
 - Order processing, customer management

Features of ERP

- Integrated system
- Real-time data processing
- Centralized database
- Automation of processes
- Improved coordination

Advantages of ERP

- Improves efficiency and productivity
- Reduces duplication of data
- Enhances decision-making
- Better resource utilization
- Improves customer satisfaction
- Streamlines operations

Disadvantages of ERP

- High implementation cost
- Complex system
- Requires training
- Time-consuming implementation
- Resistance to change

ERP Implementation Life Cycle

1. **Planning** – Define scope, goals, budget
2. **Analysis** – Study existing processes, identify gaps
3. **Design** – Configure system to fit business needs
4. **Development** – Customization and integration
5. **Testing** – UAT (User Acceptance Testing)
6. **Deployment** – Go-live
7. **Maintenance** – Ongoing support and upgrades

Real-World Example

A manufacturing company using ERP: When a customer places an order → the **Sales module** records it → **Inventory module** checks stock → **Production module** schedules manufacturing → **Finance module** generates the invoice → **Logistics module** arranges delivery. All of this happens automatically within one system.

Example of ERP (Manufacturing Industry)

Case: Automobile Manufacturing Company

A car manufacturing company uses an ERP system like SAP ERP to integrate all departments.

Without ERP:

- Production department does not know inventory status
- Sales team cannot track order progress
- Delays in communication
- Data duplication and errors

With ERP Implementation:

1.Inventory Module

- Tracks raw materials (steel, parts)
- Alerts when stock is low

2.Production Module

- Plans manufacturing schedule
- Uses real-time inventory data

3.Sales Module

- Receives customer orders
- Updates order status

4.Finance Module

- Calculates cost and revenue
- Tracks payments

5.HR Module

- Manages employee data and payroll

Working Flow:

Customer Order → Sales Module → Production Planning → Inventory Check → Manufacturing → Delivery → Payment Processing

Result:

- Reduced production delays
- Better inventory control
- Improved coordination
- Faster decision-making
- Increased customer satisfaction

Example 2: (College ERP)

A college uses ERP to manage:

- Student records
- Attendance
- Exams and results
- Fee payment

All data is stored in one system → easy access and management

Advantages of ERP

- Improves efficiency
- Reduces errors
- Provides real-time data
- Enhances coordination
- Better resource utilization

Disadvantages

- High cost
- Complex implementation
- Requires training
- Time-consuming

Supply chain management.

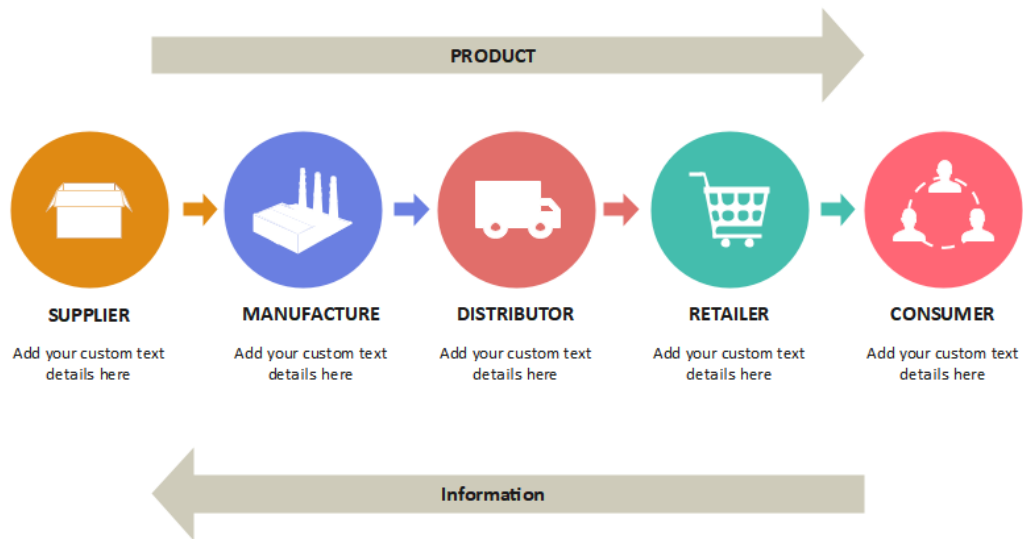
Supply Chain Management (SCM) is the management of the flow of goods, services, information, and finances from raw material suppliers to the final customer. It involves coordination among various activities such as procurement, production, inventory, transportation, and distribution.

SCM ensures that the right product reaches the right place at the right time at minimum cost.

Supply Chain Management can be defined as:

The integrated management of all activities involved in sourcing, procurement, production, and logistics to deliver products to customers efficiently.

Supply Chain Management Template Diagram



The main components include:

1. **Suppliers** – Provide raw materials
2. **Manufacturers** – Convert raw materials into products
3. **Warehouses** – Store goods
4. **Distributors** – Deliver products
5. **Retailers** – Sell to customers
6. **Customers** – Final users

Objectives of SCM

1. Ensure smooth flow of materials
2. Reduce cost and wastage
3. Improve customer satisfaction
4. Optimize inventory levels
5. Enhance coordination among departments
6. Improve delivery speed

Functions of SCM

- Procurement of raw materials
- Production planning and scheduling
- Inventory management
- Transportation and logistics
- Distribution and delivery
- Information sharing

Example of SCM (Automobile Industry)

A car manufacturing company:

- Purchases raw materials (steel, components) from suppliers
- Manufactures cars in factory
- Stores in warehouses
- Distributes to dealers
- Customers buy cars from dealers

All activities are connected through SCM for smooth flow.

Types of Supply Chains

1. Push Supply Chain
 - Production based on forecast
2. Pull Supply Chain
 - Production based on actual demand

Advantages of SCM

- Reduces operational cost
- Improves efficiency
- Faster delivery
- Better inventory control
- Enhances customer satisfaction

Disadvantages

- Complex system
- Requires coordination
- High implementation cost
- Risk of disruption

Working of SCM in Amazon / Flipkart

1.Procurement / Seller Integration

- Sellers register on platform
- Products listed online
- Real-time inventory updates

Example: A mobile seller lists products on Amazon/Flipkart

2. Warehousing (Fulfillment Centers)

- Products stored in large warehouses
- Use of automation and barcoding
- Managed by systems like Amazon's fulfillment model

Ensures quick availability of products

3. Inventory Management

- Real-time tracking of stock
- Automatic alerts for low inventory
- Demand forecasting using data

Reduces stockouts and overstock

4. Order Processing

- Customer places order online
- System selects nearest warehouse
- Order is picked, packed, and labeled

5. Transportation and Logistics

- Use of delivery partners and own logistics
- Fast movement between warehouses and hubs

Example: Amazon Transportation Services / Flipkart Ekart

6. Last-Mile Delivery

- Final delivery to customer doorstep
- Options:
 - Same-day delivery
 - Next-day delivery
 - Scheduled delivery

7. Returns Management (Reverse Logistics)

- Easy return policy
- Pickup from customer location
- Refund or replacement

Important part of e-commerce SCM

Key Features of Their SCM

- Real-time tracking
- Use of automation and AI
- Efficient warehouse management
- Fast delivery system
- Strong reverse logistics

Advantages of SCM in Amazon / Flipkart

- Fast delivery (1–2 days)
- High customer satisfaction
- Efficient inventory control
- Reduced operational cost
- Better demand forecasting

Challenges

- High logistics cost
- Managing large number of orders
- Handling returns
- Maintaining delivery speed in rural areas